

Algorithms and Pseudo-code

An algorithm is a procedure for solving a problem in terms of the actions to be executed and the order in which those actions are to be executed. It is composed of a number of steps that must be performed to solve a problem. The steps are normally "sequence," "selection," and "iteration or loop". Algorithms can be represented in number of way such as flowcharts and pseudo-code. Pseudo-code is an artificial and informal language that helps programmers develop algorithms

SEQUENCE

Sequential control is indicated by writing one action after another, each action on a line by itself, and all actions aligned with the same indent. The actions are performed in the sequence (top to bottom) that they are written.

Example (non-computer)

```
Brush teeth
Wash face
Comb hair
Smile in mirror
```

Example

```
READ height of rectangle
READ width of rectangle
area = height times width
```

SELECTION

The selection is when we want to make a choice based on some condition. the "if - else" statement.

Example (non-computer)

```
If we have any time left after swimming
    go to a restaurant
else
    go home
end
```

Example

```
If HoursWorked > NormalMax
    Display overtime message
else
    Display regular time message
end
```

ITERATION

It is used when we want the computer to keep doing something until some condition is met. The iteration is satisfied by a number of statements, such as the "while", "do", and the "for".

Example (non-computer)

```
While (kettle is not full)
    Keep adding water
end
```

Example

```
A =1
While (A < 5)
    Print A
    A = A+1
end
```

NESTED CONSTRUCTS

The constructs can be embedded within each other, and this is made clear by use of indenting. Nested constructs should be clearly indented from their surrounding constructs.

Example (XOR)

```
Input num1, num2
If (num1 is equal to 1)
    If(num2 is equal to 1)
        print 0
    else
        print 1
    end
else
    If(num2 is equal to 1)
        print 1
    else
        print 0
    end
end
```

Example

```
scholarship award = zero
marks = 0;
input marks
while (marks > 0)
    if (marks>80)
        scholarship award = scholarship award +1
    end
    input marks
end
```

Print scholarship award

In the above example, the IF construct is nested within the REPEAT construct, and therefore is indented.

More Examples:

1.

```
If (student's grade is greater than or equal to 60)
    Print "passed"
else
    Print "failed"
```

2.

```
total = zero
grade counter = one
average = 0
While (grade counter is less than or equal to ten)
    Input grade
    Add the grade into the total
average = total / ten
Print average.
```

3.

```
total = zero
counter = zero
Input grade
while (grade is not equal to sentinel (-1))
    add this grade into the running total
    add one to the grade counter
    input grade (possibly the sentinel)

if (the counter is not equal to zero)
    average = total/ counter
    print average
else
    print 'no grades were entered'
end
```

4.

```
passes = zero
failures = zero
student counter = one
while (student counter is less than or equal to ten)
```

```
input the next exam result
if (the student passed)

    add one to passes
else
    add one to failures
    add one to student counter
end
print "passes are: " passes
print "failures are: " failures
if (passes > 8)
    print "raise tuition"
end
```